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### ROLLING MACHINE FOR PRINTED SUBSTRATES AND RELATED CONTROL METHOD

#### Abstract

A rolling machine suitable for rolling up a continuous printed support, even of great length, is provided having an intermediate unit, containing a second roller with a mobile axis mounted on a movable support of the machine. The second roller is configured to receive a printed substrate and to drag it simultaneously in an direction of advancement towards the roll on which the reel is wound and in a direction perpendicular to the direction of advancement. The movable support is configured to translate perpendicularly to the movable axis of the second roller to increase/reduce a length of an overall path of the printed substrate from the printer to the reel on which it is rolled. A related control method of such a rolling machine is also disclosed.

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## Background/Summary

### TECHNICAL FIELD

[0001] The present disclosure relates to machines for rolling up printed matter and more particularly to a machine for rolling up printed substrates onto a rigid roll, such as for example large format and/or continuous sheets of paper, as they exit a printer, as well as a related control method.

### BACKGROUND

[0002] Various machines are known for rolling up on a rigid roll, for example a roll of cardboard, a substrate in the form of a continuous sheet. Such machines are used for example for the production of multiple copies of posters or other paper prints, which are reproduced in series in a continuous manner on a continuous support (for example a roll of paper) supplied to a printer. As printing progresses, a continuous support printed on at least one side comes out of the printer. This printed support must be rolled up into a roll to allow it to pass through other finishing processes before sending it to an automatic cutter, which cuts the printed media to separate individual posters or prints.

[0003] The need for such a rolling machine is particularly felt in the printing of large format continuous support, where the length and transversal size of the printed support pose severe problems for rolling it up, maintaining a certain winding tension on the reel and avoiding that it could be damaged during transport to another machine. Given the weight and bulk of the printed support to be rolled, it is crucial that it is rolled as it is printed.

### SUMMARY

[0004] Tests carried out by the Applicant have shown that a critical aspect of rolling up large format substrates as they are printed is that they must be stretched adequately when they are rolled onto the reel, but at the same time they must be allowed to exit naturally from the printer because if the substrate were pulled out of the printer while it was being printed, the print would be of unacceptable quality. Consequently, there is the problem of differentiating the mechanical tension of the printed support.

[0005] To overcome this problem, a rolling machine has been devised as defined in claim **1**. Basically, this machine has an intermediate unit **2**, comprising at least a second roller **8** with a mobile axis mounted on a movable support **9** of the machine, in which the second roller **8** is configured to receive the printed substrate **6** and to drag it simultaneously in an direction of advancement A, towards the roll on which the reel is wound and in a direction P perpendicular to the direction of advancement A, and in which the movable support **9** it is configured to translate perpendicularly to the movable axis of the second roller **8** to increase/reduce a length of an overall path of the printed substrate **6** from the printer to the reel on which it is rolled.

[0006] Also disclosed is a related method of controlling such a rolling machine, for rolling up a printed substrate, as it is produced by a printer, without any mechanical tension being exerted on the print bed and instead maintaining a desired mechanical rolling tension on a rigid roll.

[0007] Other embodiments are defined in the dependent claims.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is an overall view of a rolling machine according to a preferred embodiment of the invention.

[0009] FIG. 2 is a detail view of the input unit of the machine of FIG. 1.

[0010] FIG. 3 is a detail view of the intermediate unit of the machine of FIG. 1.

[0011] FIG. 4 is a detail view of the outlet unit and the rewind unit of the machine of FIG. 1.

[0012] FIG. 5 shows the rolling machine of FIG. 1 with the first controlled stop element raised by the first roller and with the second controlled stop element pressed against the third roller.

[0013] FIG. 6 shows the rolling machine of FIG. 1 with the first controlled stop element pressed against the first roller and with the second controlled stop element pressed against the third roller.

[0014] FIG. 7 shows the rolling machine of FIG. 1 with the first controlled stop element pressed against the first roller and with the second controlled stop element raised by the third roller.

## DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0015] The rolling machines of the present disclosure will be illustrated with reference to the preferred exemplary embodiment shown in the attached FIGS. 1 to 7, but what will be said holds also for other embodiments which do not include all the technical elements shown in the figures.

[0016] In general, a rolling machine for printed substrates of this disclosure is composed of a succession of different units aligned according to a direction of advancement A, which in the figures is a horizontal direction. More precisely, a rolling machine comprises an input unit 1, an intermediate unit 2, an output unit 3, and a rewind unit 12, in which the three units 1, 2, 3 are governed by a microprocessor unit (not shown in the figures) which manages the various electromechanical elements present in the machine. The input unit 1 has at least one first roller 4 with a fixed axis mounted on the frame 5 of the machine. The first roller 4 is configured to receive a printed substrate 6 directly at the outlet of a printer S and is configured to drag the printed substrate 6 in the direction of advancement A, as shown in FIG. 2. In the input unit 1 there is also a first controlled stop element 7 which is configured to block the printed substrate 6 against the first roller 4 to prevent it from advancing, or to allow the printed substrate 6 to advance by sliding over the first roller 4.

[0017] The intermediate unit 2, illustrated in FIG. 3, comprises at least a second roller 8, this time with a mobile axis, which is mounted on a movable support 9 of the rolling machine. The second roller 8 is configured to receive the printed substrate 6 from the input unit 1 and to drag the printed substrate 6 simultaneously in the direction of advancement A (horizontal) and in a perpendicular direction P (vertical) to the direction of advancement A. In practice, the printed substrate 6 which runs from the first roller 4 to the second roller 8 does not follow the shortest route to reach the end of the rolling machine, but in general undergoes a “deviation” in the perpendicular direction P to the direction of advancement A.

[0018] When the printed substrate 6 exits the intermediate unit 2, it is received by the output unit 3 which, as shown in FIG. 4, comprises a third roller 10 which has its axis of rotation fixed, mounted on the frame 5. The third roller 10 drags the printed substrate 6 in the direction of advancement A. Like the input unit 1, also the output unit 3 has its own controlled stop element 11 configured to block the printed substrate 6 against the third roller 10, so to prevent the printed substrate 6 from advancing, or to allow the printed substrate 6 to be dragged by the third roller 10 towards the final part of the rolling machine.

[0019] Downstream of the outlet unit 3 in the forward direction A, the rolling machine comprises a rewind unit 12 also mounted on the frame 5 of the machine. The rewind unit 12 is configured to rewind the printed substrate 6 proceeding from the output unit 3 and comprises a motorized shaft 13 configured to be inserted into a rigid roll, such as a roll of cardboard, which is rotated such that the printed substrate 6 is rolled up on it. As shown in FIG. 1, the various units 1, 2, 3, 12 are arranged along the direction of advancement A of the rolling machine, from the input unit 1 to the

rewind unit **12**. As shown in FIG. **1**, the intermediate unit **2** has the movable support **9** which allows at least the second roller **8** to move at least partially in the direction P perpendicular to the direction of advancement A and perpendicular to the movable axis of the second roller **8**, so as to increase or reduce an overall distance of the second roller **8** with respect to the first roller **4** and the third roller **10**, to increase or reduce an overall path of the printed substrate **6** which goes from the first roller **4** to the third roller **10**.

[0020] In the embodiment illustrated in FIG. **1**, reference is made to the preferred case in which the intermediate unit **2** also comprises a fourth transmission roller **14** with a fixed axis, mounted on the frame **5** of the machine, configured to receive the printed substrate **6** from the second roller **8**, as well as a fifth roller **15** with a mobile axis, also mounted on the movable support **9** like the second roller **8**, also configured to translate perpendicularly to its mobile axis.

[0021] The fifth roller **15** is configured to receive the printed substrate **6** from the fourth roller **14** with a fixed axis and to drag it simultaneously in the forward direction A and in the direction P perpendicular thereto, thanks to the movement of the movable support **9** on which the fifth roller **15** is mounted. In this case, the third roller **10** with fixed axis of the output unit **3** receives the printed substrate **6** not directly from the second roller **8**, but from the fifth roller **15** with mobile axis.

[0022] According to an optional aspect, the output unit **3** comprises a sixth idle roller **16** with a fixed axis mounted on the frame **5** of the machine and configured to receive the printed substrate **6** from the third roller **10**. In this case, the winding unit **12** it is configured to rewind the printed substrate **6** which is supplied by the sixth idle roller **16** instead of by the third roller **10**.

[0023] According to one aspect of the rolling machine of this disclosure, the microprocessor unit not shown in the figures is configured to control the controlled stop element **7** of the input unit **1** so as to block the printed substrate **6** by squeezing it against the first roller **4** when the printed substrate **6** is to be stretched. In this way, the printed substrate **6** is not pulled by the intermediate unit **2** and this prevents the support **6** from moving while the printing is in progress, which would lead to a qualitatively unacceptable printing.

[0024] When the controlled stop element **7** blocks the printed substrate **6**, the rewind unit **12** can continue to rewind the printed substrate **6** because the movable support **9** of the intermediate unit **2** is moved in the perpendicular direction P so as to shorten the overall path of the printed substrate **6** from the first roller **4** to the rewind unit **12**.

[0025] On the other hand, when the printed substrate **6** is pushed out of the printer S and the intermediate unit is ready to accept it, then the controlled stop element **7** is configured so as not to block the printed substrate **6** anymore, but to allow it to be dragged from the first roller **4**. If instead for any reason the rewinding of the printed substrate **6** on the rigid roll in the rewind unit **12** should be stopped, then the second controlled stop element **11** is configured so as to press onto the printed substrate **6** to block it against the third roller **10** and thus maintain a tension. In this circumstance, the movable support **9** of the intermediate unit **2** is lowered along the perpendicular direction P so as to increase the overall path of the printed substrate **6** from the first roller **4** of the input unit **1** to the third roller **10** of the output unit **3**.

[0026] According to one aspect, each one of the controlled stop elements **7** and **11** comprises a respective movable bar configured to be pressed on the printed substrate **6** respectively against the first roller **4** and against the third roller **10**, when it is desired to prevent the printed substrate **6** from advancing, and to be moved away respectively from the first roller **4** and from the third roller **10** when one wishes instead to allow the printed substrate **6** to advance in the forward direction A towards the rewind unit **12**.

[0027] According to one aspect, shown in the figures, the respective movable bars of the controlled stop elements **7**, **11** are mounted on respective arms **17**, **18** hinged to the frame **5** of the rolling machine.

[0028] According to one aspect, the movable support **9** of the intermediate unit **2** is made in the form of an arm **19** hinged to the frame **5** of the machine so that, when the hinged arm **19** rotates,

the mobile axis of the second roller **8** and, if present, also of the fifth roller **15**, moves at least partially along the perpendicular direction **P** so as to increase or reduce the overall path that the printed substrate **6** takes inside the intermediate unit **2**. As an alternative, not shown in the figures, the movable support **9** can be a part of the rolling machine which moves along the perpendicular direction **P** with an upward or downward movement.

[0029] In one aspect, all the rollers (or roller) located in the intermediate unit **2** are non-driven idle rollers.

[0030] According to one aspect, not shown in the figures, the movable support **9** can be moved/counterbalanced by an electromechanical actuator (not shown in the figure), in turn controlled by the microprocessor unit which also controls the controlled stop elements **7**, **11**.

[0031] The winding machine of this disclosure allows to rewind any type of printed support without exerting any tension on the support itself during the printing phase, while still maintaining a good rewinding tension of the reel and a satisfactory lateral alignment. This excellent result is obtained by controlling the described rolling machine in the following manner.

[0032] Initially, the controlled stop elements **7**, **11** are raised, then the printed substrate passes over the rollers of input unit **1**, through the rollers of intermediate unit **2** and of output unit **3** and the printed support **6** is fixed to the rigid roll inserted on the motorized shaft **13** of the rewind unit **12**. The printed support **6** is blocked against the third roller **10** by lowering the second controlled stop element **11**, then the movable support **9** is lowered so as to increase the length of the overall path of the printed substrate **6** from the first roller **4** to the third roller **10**, as illustrated in FIG. 5. In this phase of the method, the motorized shaft **13** of the rewind unit **12** is stationary because the printed substrate **6** is stopped against the third roller **10**, and the movable support **9**, by lowering, begins to feed the intermediate unit **2** and then to tense the substrate.

[0033] As soon as the printed substrate **6** begins to be taut, the substrate is locked against the first roller **4** through the first controlled stop element **7**, as shown in FIG. 6. In this configuration, the input unit **1**, the intermediate unit **2** and the output unit **3** are separate and this makes possible to avoid pulling the printed substrate **6** from the printer **S**, which can therefore continue printing without tension being applied to the substrate on the printing plane, and the part of printed substrate **6** which goes from the third roller **10** to the winding unit **12** is kept taut. Meanwhile, the movable support **9**, thanks to the fact that it can move at least partially in the perpendicular direction **P** and under the action of its own weight, tenses the printed substrate **6** which goes from the first roller **4** to the third roller **10**.

[0034] Once a sufficient tension to roll the printed substrate **6** onto the reel is obtained, the microprocessor unit of the rolling machine controls the second element **11** so as to allow the printed substrate **6** to advance by sliding on the third roller **10**, as shown in the FIG. 7. The printed substrate **6** is then wound with a desired tension which is not transmitted to the printing plane of the printer **S** because the first controlled stop element **7** holds the printed substrate **6** still by pressing it against the first roller **4**. As the substrate printed substrate **6** is wound on the reel of the rewind unit **12**, the movable support **9** moves at least partially along the perpendicular direction **P** to the direction of advancement **A** so as to reduce the length of the overall path of the printed substrate from the first roller **4** to the third roller **10**.

[0035] After winding the printed substrate **6** on the reel to the point of having reduced the overall path of the printed substrate **6** from the first roller **4** to the third roller **10** up to a minimum nominal length, the microprocessor unit commands the second controlled stop element **11** (FIG. 6) so as to keep under tension the portion of printed substrate **6** which goes from the third roller **10** to the rewind unit **12**. The first element **7** is then actuated with a stop command so that the printed substrate **6**, which has come out of the printer **S** while the rewind unit **12** was rolling it up on the reel, can slide on the first roller **4** and enter the intermediate unit **2**, as illustrated in FIG. 5. In this phase the movable support **9** will move at least partially along the perpendicular direction **P** so as to increase the length of the overall path of the printed substrate **6** from the first roller **4** to the third

roller **10**. At this point, the process continues cyclically as previously indicated with reference to FIGS. **6** and **7** until the reel is completely rewound.

[0036] When the printed support **6** ends, the machine waits for the intervention of an operator to roll up the final part of the support and to remove the reel from the motorized shaft **13**. According to one aspect, the motorized shaft **13** is a shaft of the type with expandable strips, on which it is possible to insert a rigid roll, for example of cardboard, which becomes integral with the motorized shaft **13** thanks to the expansion of the strips.

[0037] In the embodiments shown in the figures, the printed substrate **6** is taut in the desired way (FIGS. **6** and **7**) thanks to the weight of the movable support **9** itself. By suitably adjusting the weight of the movable support **9**, possibly counterbalanced by an **5** electromechanical actuator also controlled by the microprocessor unit, it will be possible to adjust the mechanical tension applied to the printed support **6** to be rolled up and it will also be possible to control the movement speed of the movable support **9** in the perpendicular direction P, which in the embodiment illustrated in the figures is a descent speed, so that the printed support **6** enters the intermediate unit **2** without generating **10** unacceptable jerks on the support part in the printing plane of the printer S.

## Claims

**1.** A rolling machine for printed substrates, comprising in succession in a direction of advancement: an input unit, comprising: a first fixed-axis roller mounted on a frame of said machine, configured to receive a printed substrate directly at the output of a printer and to drag said printed substrate in the direction of advancement, a first controlled stop element, configured to lock the printed substrate against the first roller and to prevent the printed substrate from advancing, or to allow the printed substrate to advance by sliding on the first roller, an intermediate unit, comprising at least a second roller with movable axis mounted on a movable support of said machine, said second roller being configured to receive said printed substrate from the input unit and for dragging the printed substrate simultaneously in the direction of advancement and in a direction perpendicular to the direction of advancement; an output unit, comprising: a third roller with a fixed axis mounted on said frame, configured to receive said printed substrate from the intermediate unit and to drag the printed substrate in the direction of advancement, a second controlled stop element, configured to lock the printed substrate against the third roller and to prevent the printed substrate from advancing, or to allow the printed substrate to advance by sliding on the third roller, a rewind unit mounted on a frame of said machine, the rewind unit being configured to rewind the printed substrate that advances from the output unit and comprising a motorized shaft configured to be inserted into a rigid roll and to rotate it to roll-up the printed substrate onto the rigid roll; a microprocessor unit configured to: control said first controlled stop element to lock the printed substrate against the first roller when the printed substrate in the intermediate unit is to be made tense, controlling said second controlled stop element so as to lock the printed substrate against the third roller while the printed substrate is not rewound on the rigid roll; wherein said movable support is configured to translate perpendicularly to the movable axis of the second roller to increase/reduce a total distance of said second roller from said first roller and from said third roller, so as to increase/reduce a length of an overall path of said printed substrate from the first roller to the third roller.

**2.** The machine according to claim 1, wherein each one of said first controlled stop element and said second controlled stop element comprises a respective movable bar configured to be pressed on the printed substrate respectively against the first roller and against the third roller, to prevent the printed substrate from advancing, and to be moved away from the first roller and from the third roller respectively, to allow the printed substrate to advance.

**3.** The machine according to claim 2, wherein each respective movable bar of said first stop controlled element and of said second stop controlled element is mounted on a respective arm

hinged to the frame of said machine.

**4.** The machine according to claim 1, wherein said movable support on which the second roller with movable axis is mounted is realized in the form of a rotating arm hinged to the frame of the machine so that, when the rotating arm rotates, the mobile axis of the second roller moves at least partially along said perpendicular direction.

**5.** The machine according to claim 1, wherein: said intermediate unit comprises: a fourth return roller with fixed axis mounted on the frame of the machine, configured to receive said printed substrate from the second roller, a fifth roller with movable axis mounted on said movable support, configured to translate perpendicularly to the movable axis of the fifth roller, said second roller being configured to receive said printed substrate from the fourth roller and to drag the printed substrate simultaneously in the direction of advancement and in said perpendicular direction; said third roller with fixed axis being configured to receive said printed substrate from the fifth roller.

**6.** The machine according to claim 1 wherein: said output unit comprises a sixth idle roller with fixed axis mounted on said frame and configured to receive said printed substrate from the third roller; said rewind unit being configured to rewind the printed substrate which advances from said sixth idle roller.

**7.** The machine according to claim 1, in which all the rollers of said intermediate unit are idle rollers.

**8.** The machine according to claim 1, further comprising an electromechanical actuator controlled by said microprocessor unit, configured to move/counterbalance said movable support.

**9.** A method of controlling a rolling machine of a printed substrate according to claim 1, comprising: controlling, with said microprocessor unit, said first controlled stop element to lock the printed substrate against the first roller when the printed substrate in the intermediate unit is to be made tense; controlling, with said second microprocessor unit, said second controlled stop element to block the printed substrate against the third roller while the printed substrate is not rewound on a rigid roll; translating said movable support perpendicularly to the movable axis of the second roller to increase/reduce an overall distance of said second roller from said first roller and from said third roller, to increase/reduce a length of an overall path of said printed substrate from the first roller to the third roller.

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